

# ECE284 FA25 Final Progress Report

1. Fill the table below. This helps us to understand your current progress status.

\*Model Issue: A group member got on plane without sharing model model\_best.pth.tar and checkpoint.pth. Might have to submit it after submission deadline. Our ipynb cell output has demonstrated that to prove that the model actually works. We will submit it ASAP once our teammate lands.

| Item                                         | Current Status                  | Status during Poster Presentation | Note                                                                                                                                      |
|----------------------------------------------|---------------------------------|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Part1                                        | Complete                        | Complete                          | *Model Issue                                                                                                                              |
| Part2                                        | Complete                        | Complete                          | Using debug data, software data doesn't match. But we are pretty confident about the hardware design should be right.<br><br>*Model Issue |
| Part3                                        | Complete                        | Complete                          | *Model Issue                                                                                                                              |
| Alpha 1 (Hybrid Pruning)                     | Complete/ *software checkpoints | Complete                          | Perform unstructured and structured pruning together to reach a better accuracy and preserve regularity                                   |
| Alpha 2 (Approx. Multiplier for Low Power)   | Complete                        | Complete                          | Replace multiplication with approximate version to reduce power of MAC while inducing negligible error.                                   |
| Alpha 3 (Region-aware Activation Extraction) | Complete                        | Complete                          | Selecting necessary features for each Kij in the kernel to reduce unnecessary psum computation.                                           |

|                                                |          |          |                                                                                                                                                                            |
|------------------------------------------------|----------|----------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Alpha 4 (Hybrid Read Mode and Pipelined RD/WR) | Complete | Complete | Allowing FIFO to select different read signal propagation for each row of FIFO and also supporting pipelined Read/Write operation to minimize data communication overhead. |
| Alpha 5 (Zero Skipping MAC Operation)          | Complete | Complete | Gating the FF to stop data changes when weight or input are zero.                                                                                                          |